

GraphTrust Publication Figure Set

Baktiyar Ablaihan · NSRI Summer Research Hackathon 2026

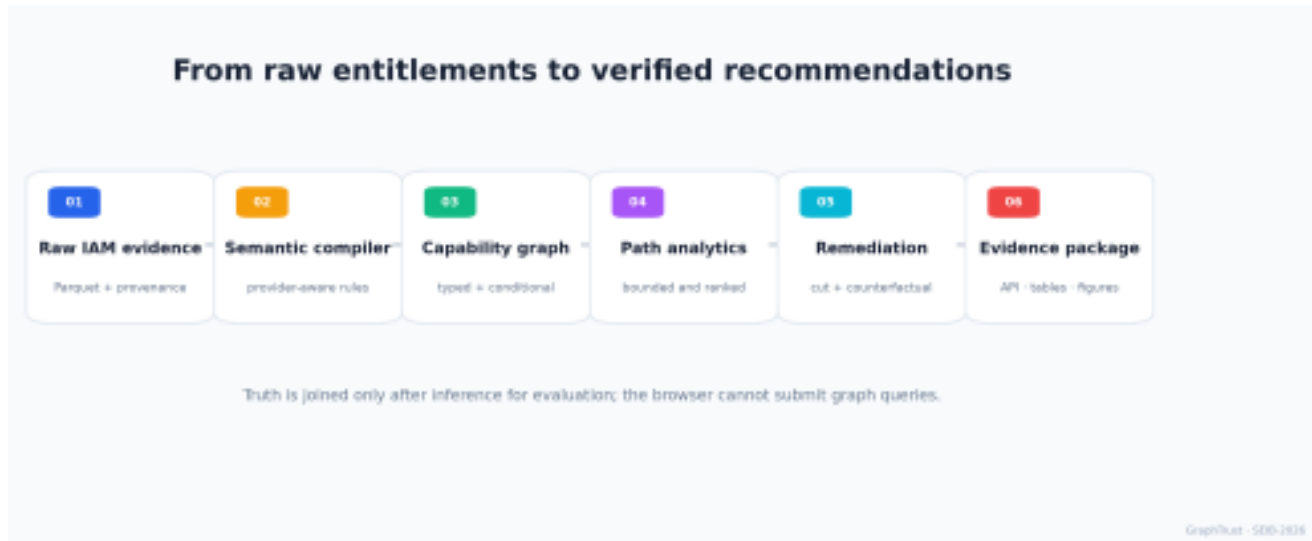


Figure 1: Evidence-preserving GraphTrust analysis pipeline.

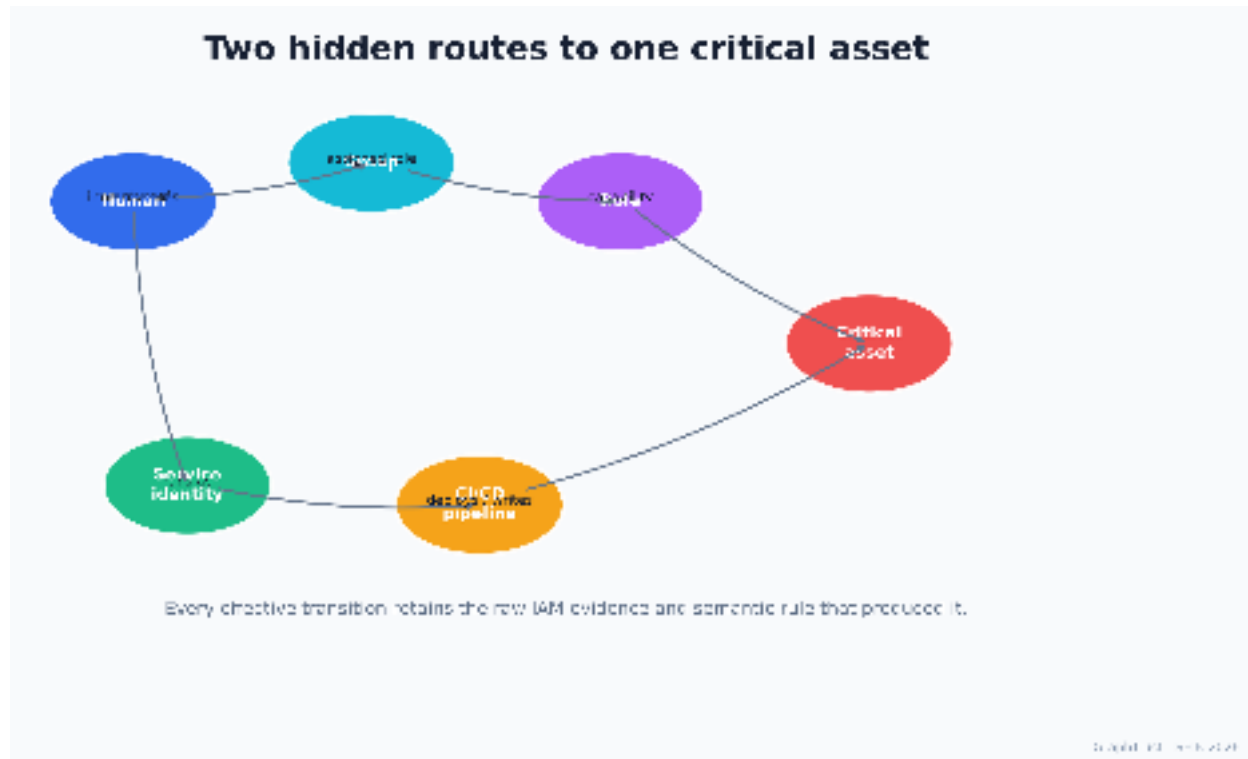


Figure 2: Canonical heterogeneous identity graph and two transitive routes.



Figure 3: Highest-ranked evidence-linked privilege path.

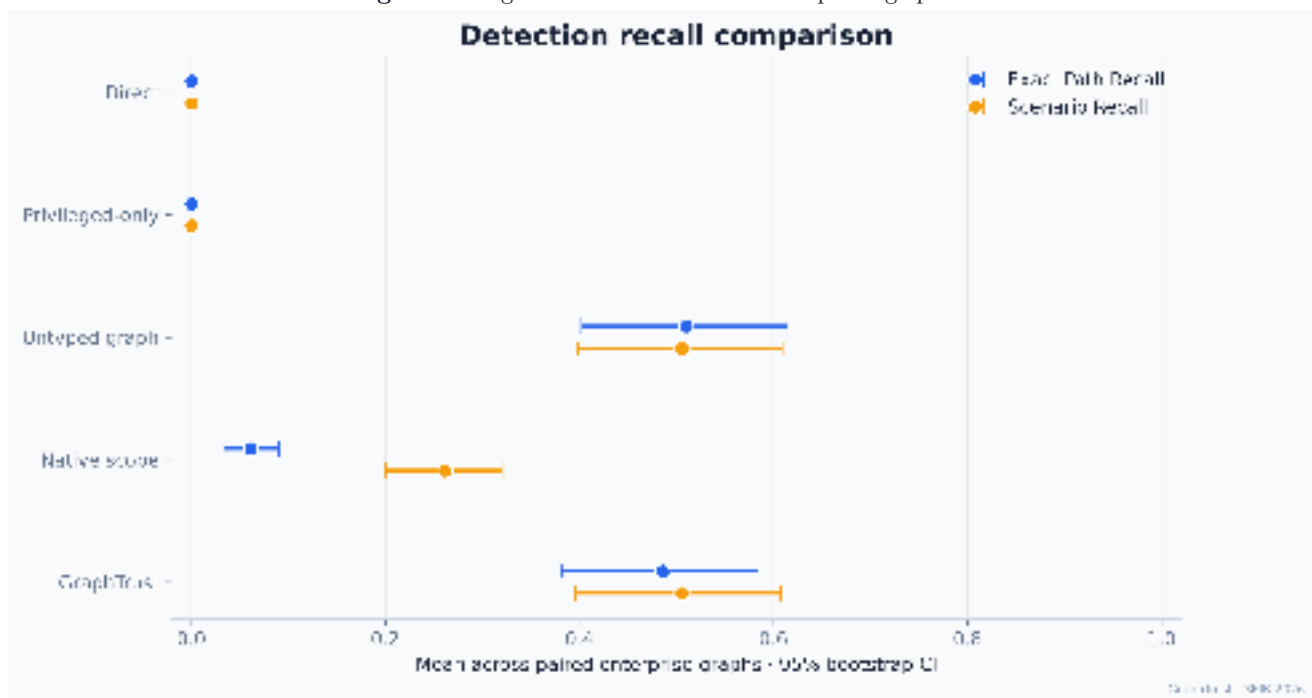


Figure 4: Detection recall with graph-level bootstrap intervals.

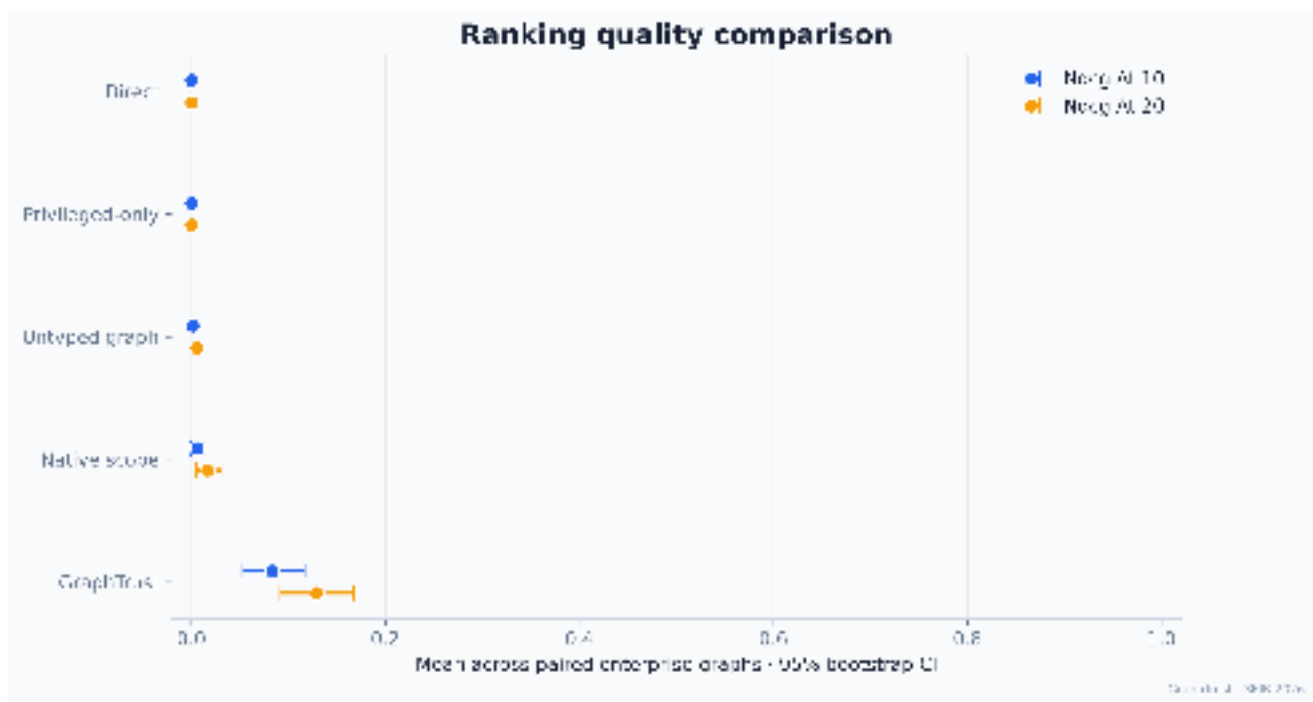


Figure 5: Top-ranked relevance across the five audit methods.

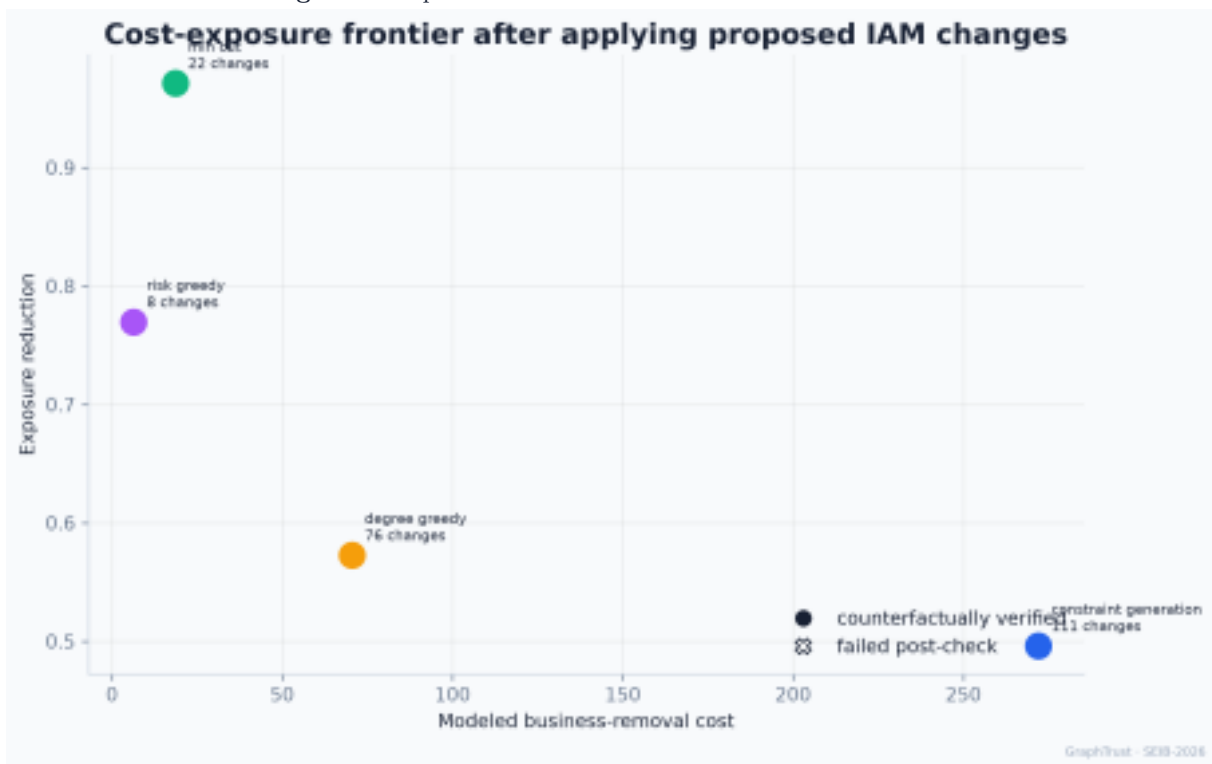


Figure 6: Remediation cost, exposure reduction, and post-check state.

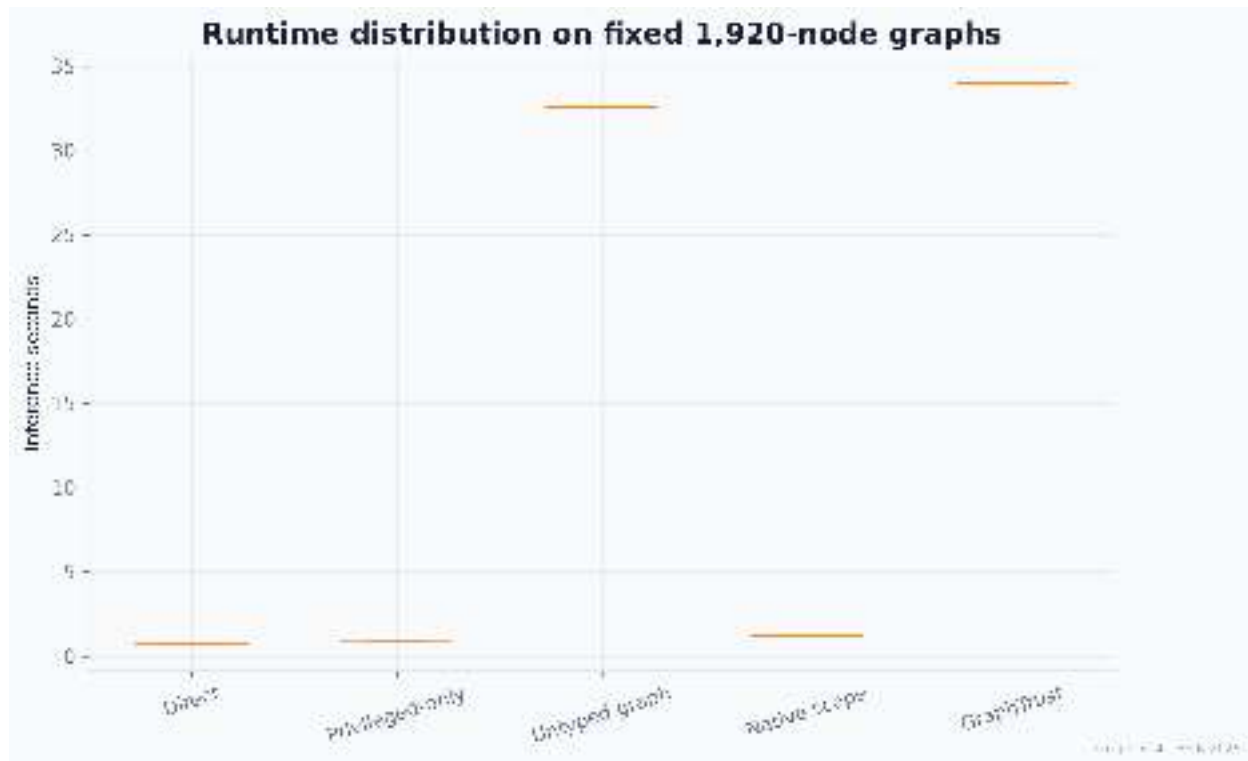


Figure 7: Runtime distribution on fixed-size graphs; no scaling claim is made.



Figure 8: GraphTrust ZTRI distributions across enterprise profiles.

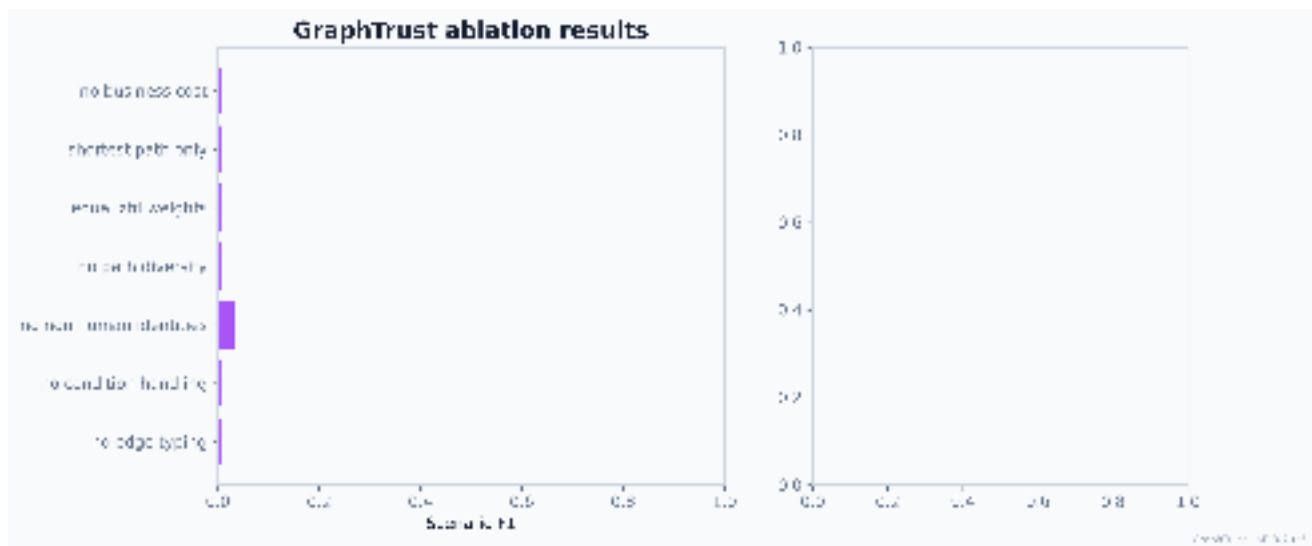


Figure 9: Ablation study on the pre-specified verification graph.



Figure 10: One-at-a-time and 1,000-sample weight sensitivity.

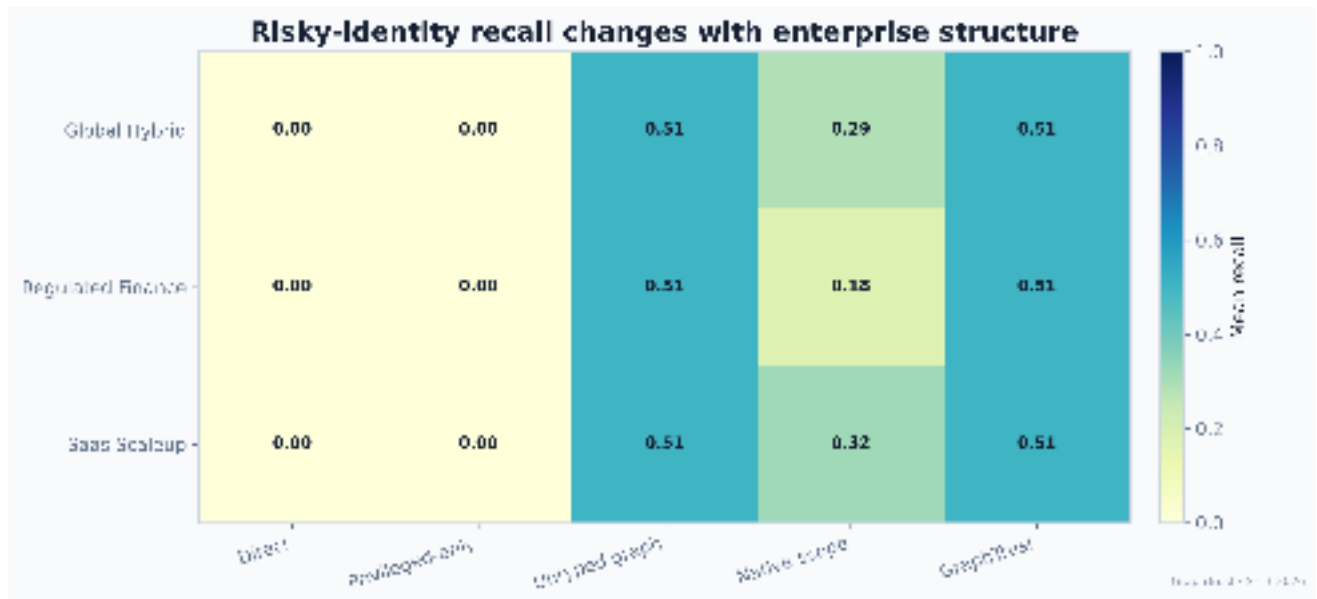


Figure 11: Risky-identity recall by enterprise structure.

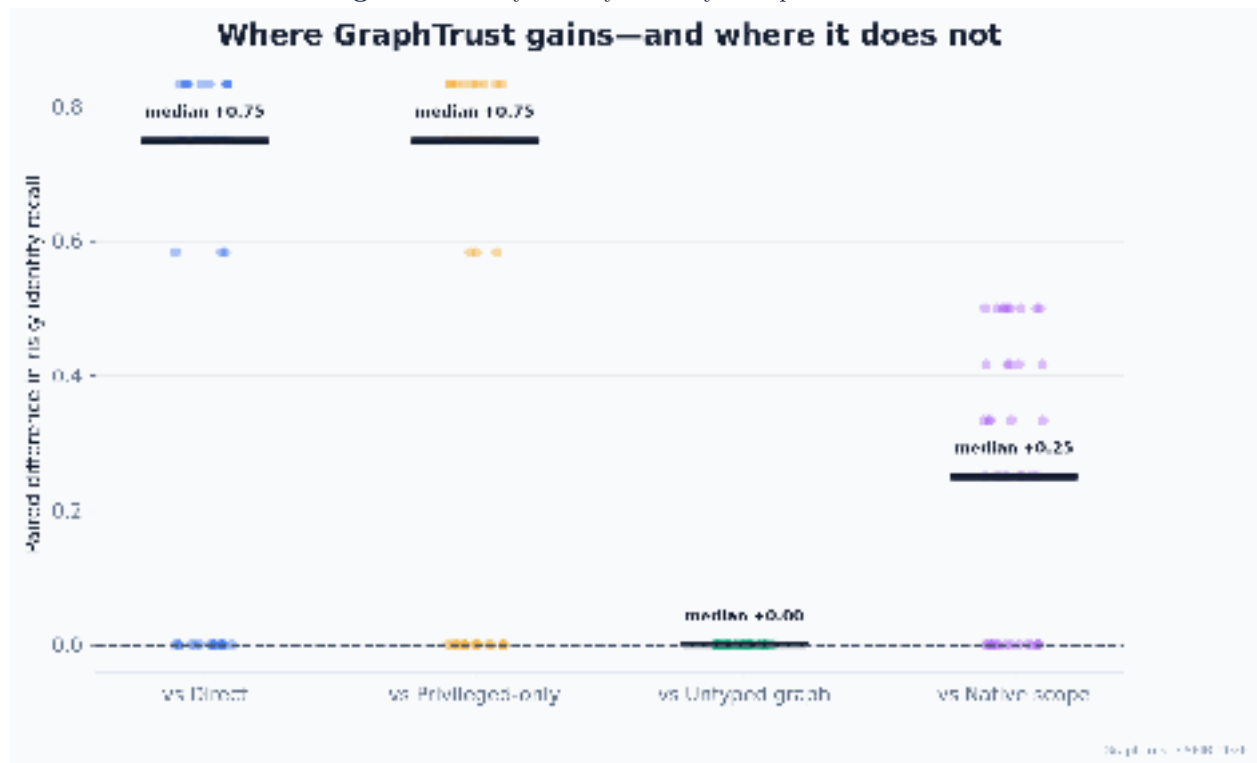


Figure 12: Paired GraphTrust differences against each baseline.

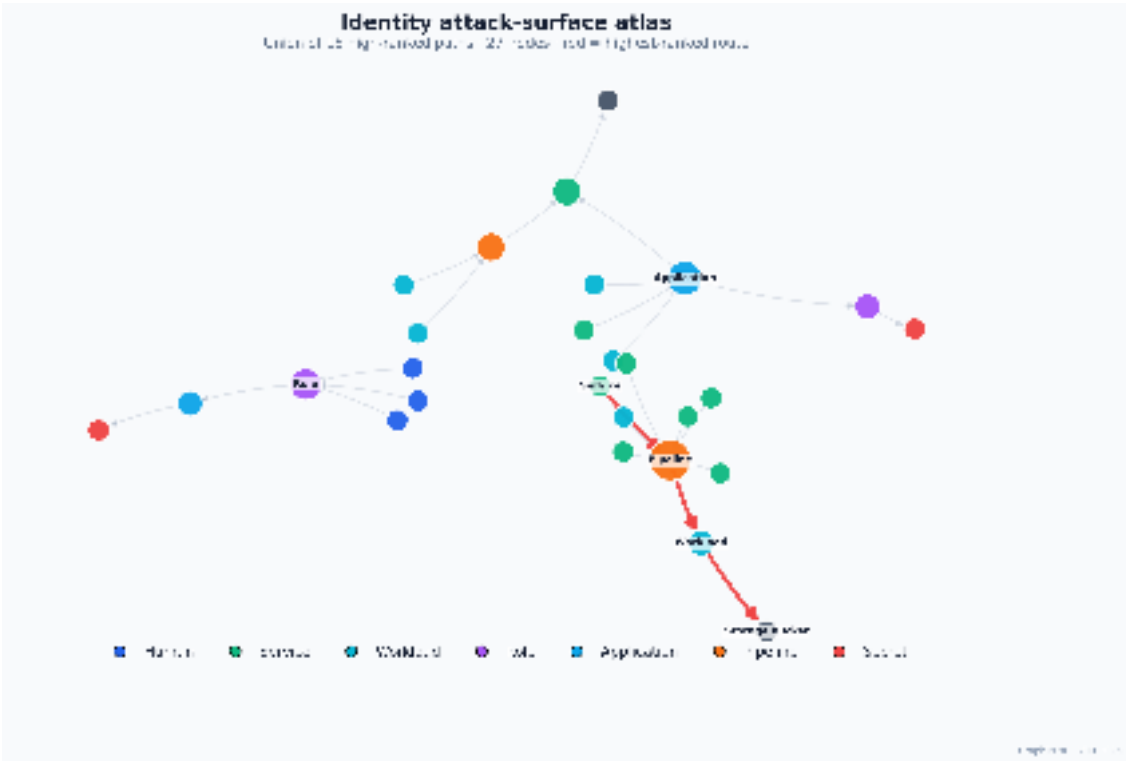


Figure 13: Evidence-backed attack-surface atlas with the highest-ranked route highlighted.

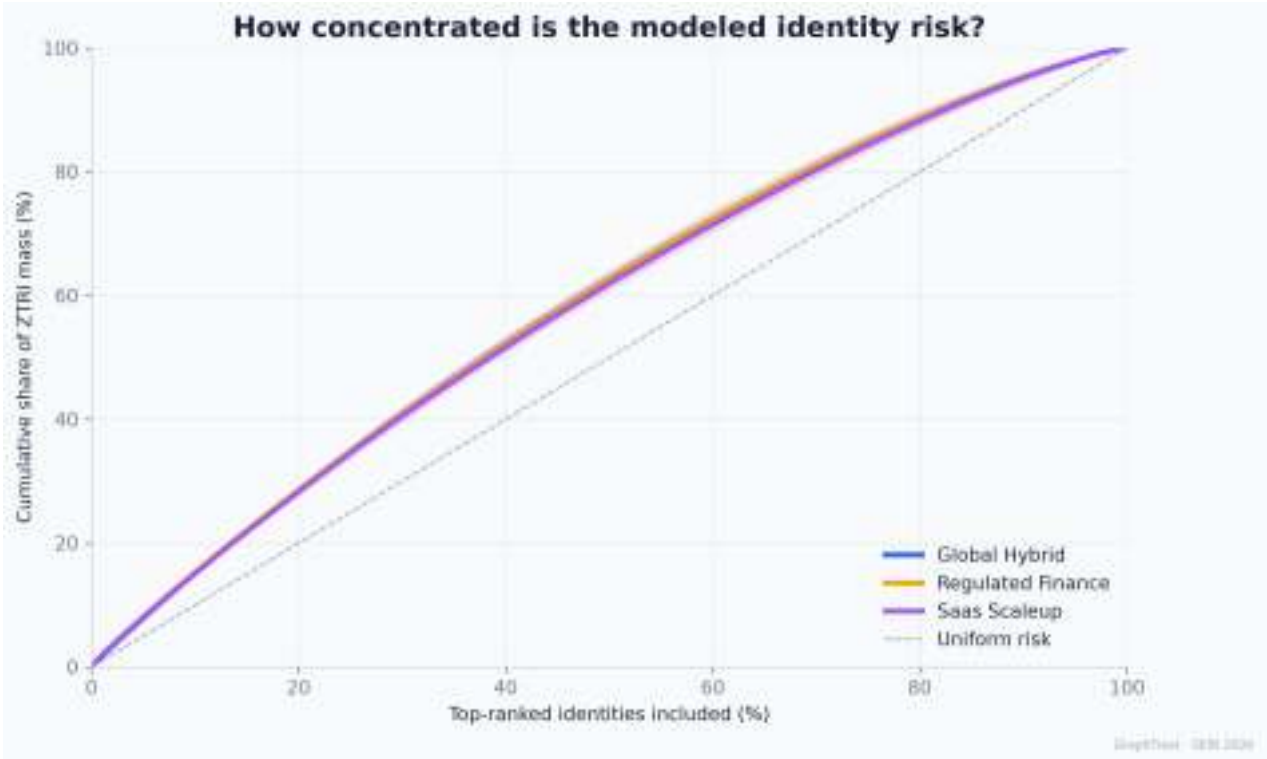


Figure 14: Concentration of modeled identity risk across enterprise profiles.

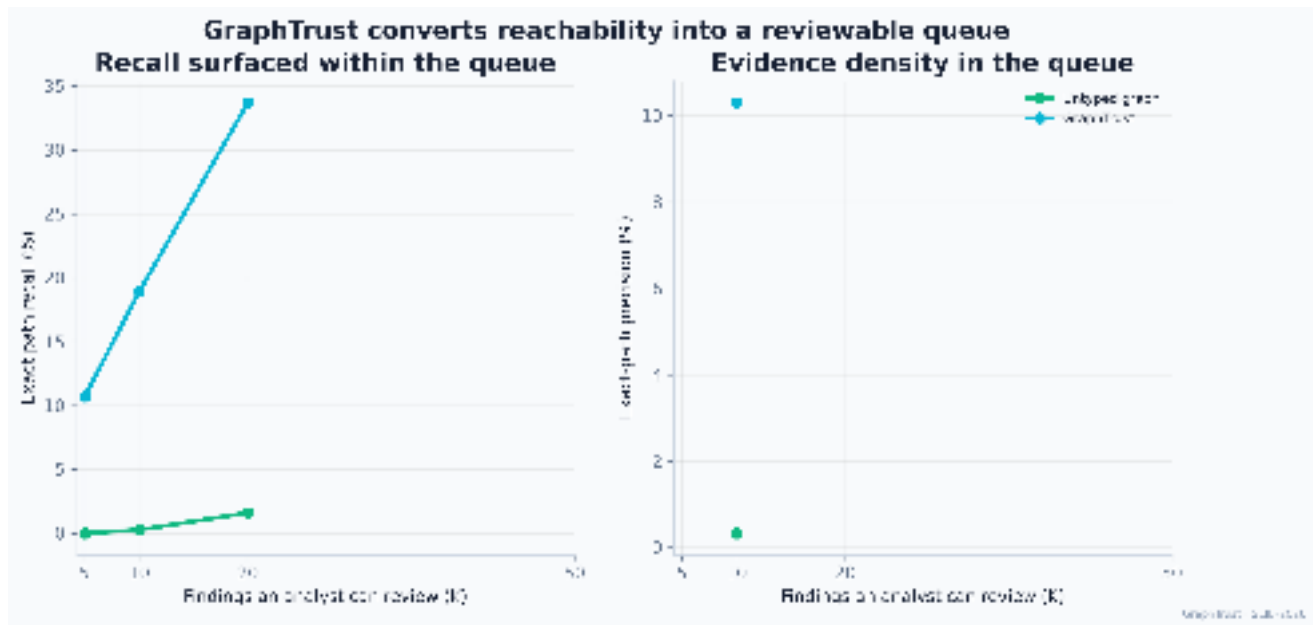
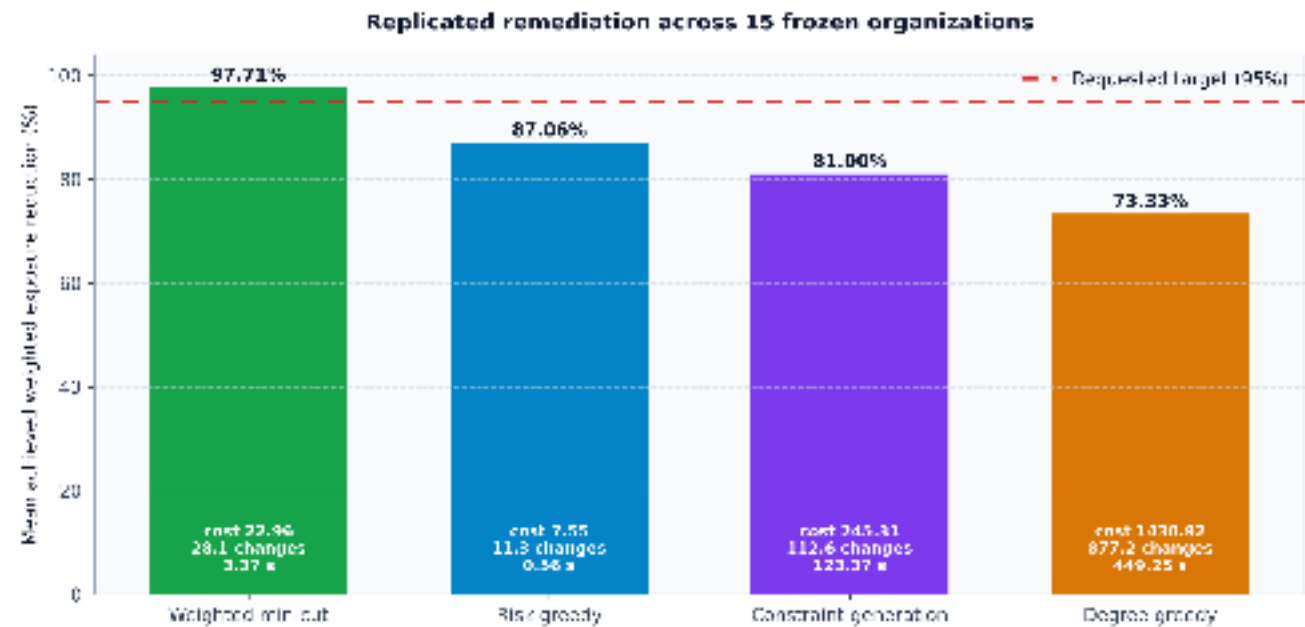
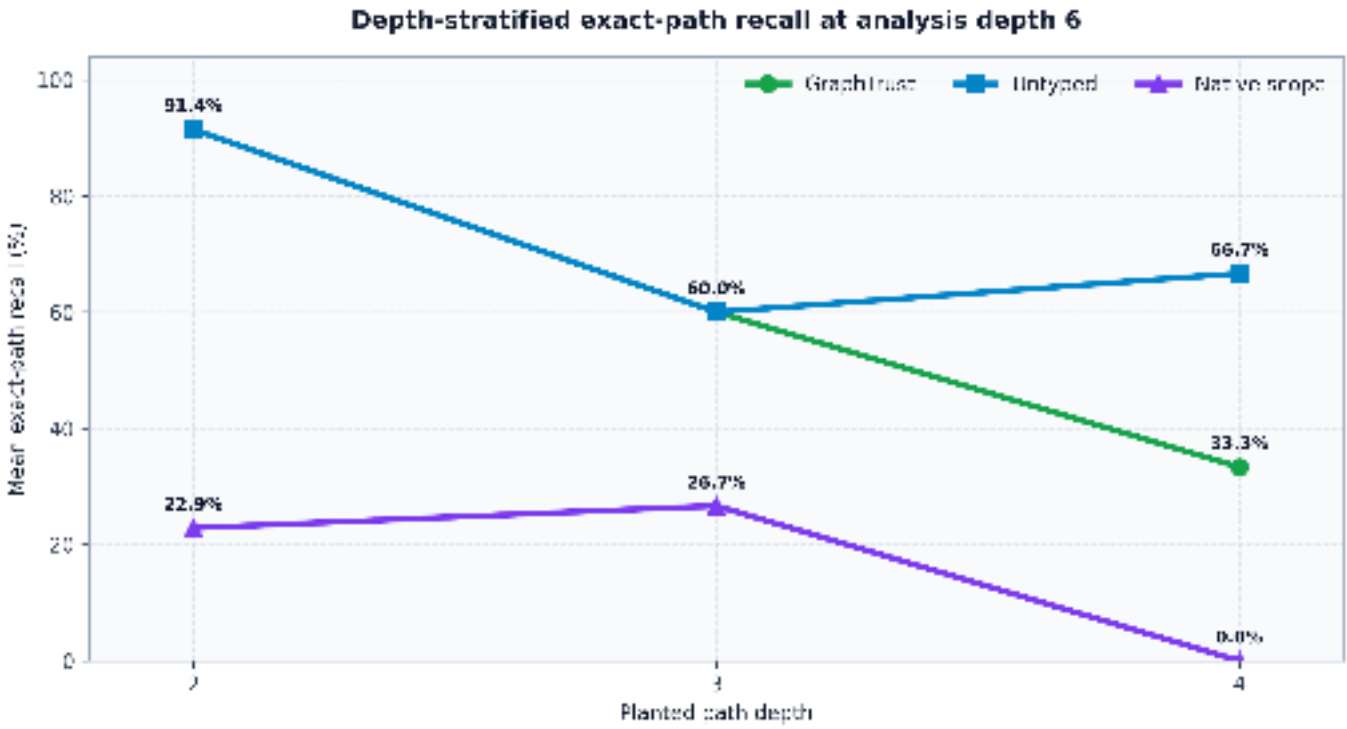


Figure 15: Analyst review-budget curve from the operator audit reanalysis.



All counterfactuals and worlds verified. Requested target attainment is reported separately.

Figure 16: Replicated remediation outcomes across 15 frozen organizations.



n = 15 independent organizations. Direct and privileged-only recall = 0; no depth-0 planted keys exist.

Figure 17: Depth-stratified exact-path recall at analysis depth six.